

GND IT TESTING FACILITY

SHORT-CIRCUIT TRIP VALIDATION REPORT

Report No: GND-20260924-220621 | Lead Engineer: Krishna Rakesh Kumar Jaiswar

TEST SPECIFICATION

Date: 2026-09-24 22:06:21

DUT ID: MCB-001

DUT Config: Single Pole (1P)

DUT Rating: 32 A, curve C, Icn 6 kA

Voltage Source: 3.0 Vrms @ 50 Hz

Target Isc (RMS): 1000 A

Target cos ϕ : 0.5

Closing Angle α : 0°

HARDWARE & SYNTHESIZED CIRCUIT

Bank Code: R=0b000110 / L=0b001100

Loop R / L: 1.50 m Ω / 8.0 μ H

Synthesized Isc / cos ϕ : 1025 A / 0.512

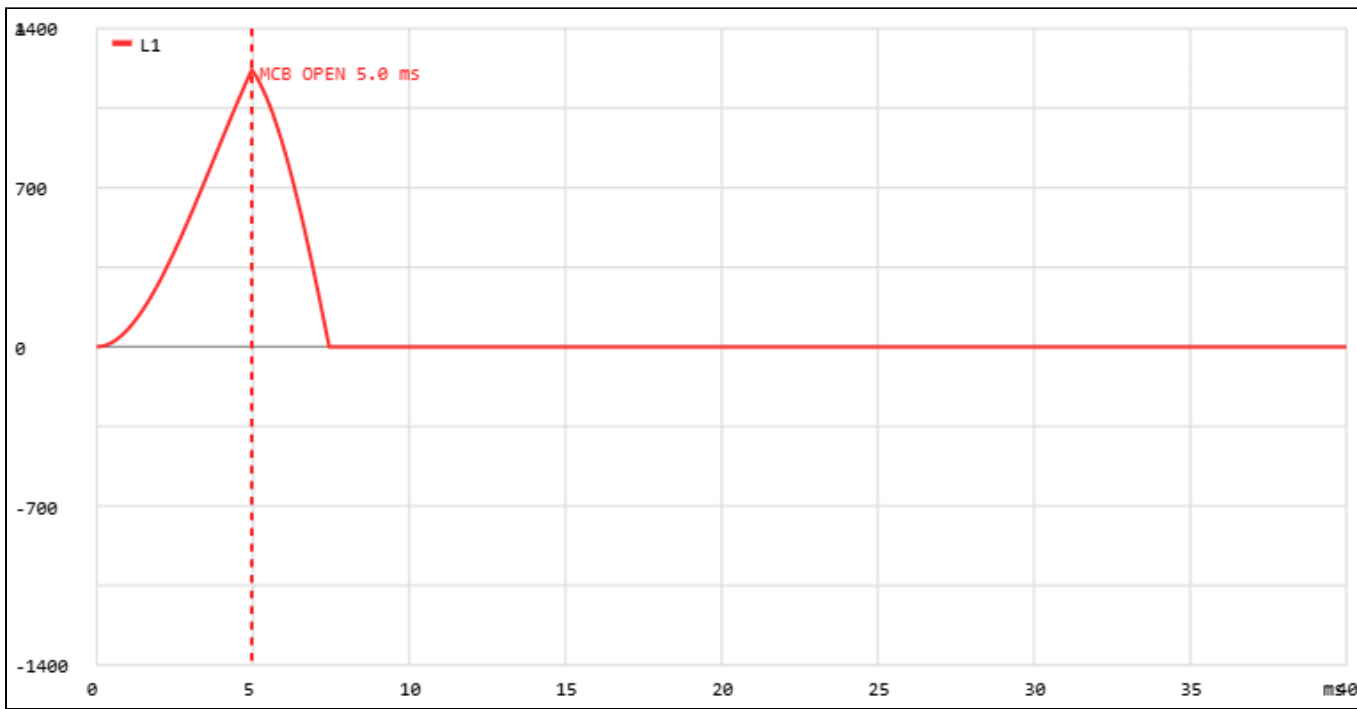
Clamping: Pneumatic block (6.8 bar)

Switching Element: Multi-channel SCR sync

Data Acquisition: 100 kHz, 4000 samples/ch

Bank Temp (at shot): 32.0 °C (blower active)

CAPTURED WAVEFORM



POLE / PHASE TELEMETRY & RESULTS

POLE	PEAK CURRENT (IP)	LET-THROUGH (I ² T)	CONTACT OPENING	TOTAL CLEARING	VERDICT
L1	1218 A	3476 A ² s	5.0 ms	7.4 ms	PASS

TRIP ANALYSIS

Isc/In = 32.0x. Curve C magnetic band = 160 - 320 A. Result: ALL POLES CLEARED (GANGED). I²t acceptance limit: 100,000 A²s.

DUT TRIP TEST: PASS

Notes: Simulated bench record. I_{sc} is the RMS symmetrical prospective current synthesized at 3.0 V RMS through the binary R-L bank including source impedance. I_p and I^2t are computed from 100 kHz samples of each pole channel. This bench verifies magnetic (instantaneous) tripping up to 1.5 kA; it is not an I_{cn} / I_{cs} breaking-capacity certification under IEC 60898-1:2015, which requires the rated fault level in an accredited high-power laboratory.

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Reviewed / Approved by: